

Entropy of example dataset

- Situations where I will/won't wait for a table:

Example	Attributes										Target
	<i>Alt</i>	<i>Bar</i>	<i>Fri</i>	<i>Hun</i>	<i>Pat</i>	<i>Price</i>	<i>Rain</i>	<i>Res</i>	<i>Type</i>	<i>Est</i>	<i>Wait</i>
X_1	T	F	F	T	Some	\$\$\$	F	T	French	0-10	T
X_2	T	F	F	T	Full	\$	F	F	Thai	30-60	F
X_3	F	T	F	F	Some	\$	F	F	Burger	0-10	T
X_4	T	F	T	T	Full	\$	F	F	Thai	10-30	T
X_5	T	F	T	F	Full	\$\$\$	F	T	French	>60	F
X_6	F	T	F	T	Some	\$\$	T	T	Italian	0-10	T
X_7	F	T	F	F	None	\$	T	F	Burger	0-10	F
X_8	F	F	F	T	Some	\$\$	T	T	Thai	0-10	T
X_9	F	T	T	F	Full	\$	T	F	Burger	>60	F
X_{10}	T	T	T	T	Full	\$\$\$	F	T	Italian	10-30	F
X_{11}	F	F	F	F	None	\$	F	F	Thai	0-10	F
X_{12}	T	T	T	T	Full	\$	F	F	Burger	30-60	T

12 examples

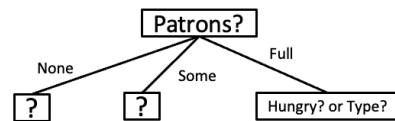
6 +

6 -

- What's the entropy of this collection of examples?
- $p = 6; n = 6; I(6,6) = -0.5 \log(0.5) - 0.5 \log(0.5) = 1$
- Need 1 bit of information to classify a randomly picked example from this data

Classwork

- Compute whether *Hungry* or *Type* is the better attribute after *Patrons* is made the root



Group	Wait	Entropy
Hungry = T	(2,2)	1
Hungry = F	(0,2)	0

$$\text{Expected}(\text{Hungry}) = \frac{4}{6} (1.0) + \frac{2}{6} (0.0) = 0.667$$

Group	Wait	Entropy
Thai	1,1	1
Burger	1,1	1
French	0,1	0
Italian	0,1	0

$$\text{Expected(Type)} = \frac{2}{6} (1.0) + \frac{2}{6} (1.0) + \frac{1}{6} (0.0) + \frac{1}{6} (0.0) = 0.667$$

So they are the same so patron is the better split

Class work

- For the dataset
- Calculate the
 - Gini index of $X < 1.35$ ("left")
 - Gini index of $X > 1.35$ ("right")
 - $J(X, 1.35)$
- Repeat for $X < 2.4$ and $X > 2.4$
- Is it better to split at $X=1.35$ or $X=2.4$?

X	target
1.2	N
1.5	Y
1.7	Y
2.3	Y
2.5	N
3.0	N

Goal. Compare two threshold splits using Gini, combine via the weighted criterion J, pick the lower J.

Gini: $G = 1 - \sum p(c_i)^2$ $G = 1 - \sum p(c_i)^2$ (square each probability — mandatory; the squares are what make Gini informative)

Split criterion: $J(X, t) = m_{\text{left}} G_{\text{left}} + m_{\text{right}} G_{\text{right}}$ $J(X, t) = \frac{m_{\text{left}}}{m} G_{\text{left}} + \frac{m_{\text{right}}}{m} G_{\text{right}}$ $J(X, t) = m_{\text{left}} G_{\text{left}} + m_{\text{right}} G_{\text{right}}$ (**lower J = purer = better**; no "parent minus" step — compare J's directly)

Split at X = 2.4

Cut sits *after* the Y-block, so the Y's land on the **left** this time (opposite side from 1.35).

Side	Rows	Counts	Gini
Left ($X < 2.4$)	1.2, 1.5, 1.7, 2.3	3 Y, 1 N	$1 - (0.75^2 + 0.25^2) = \mathbf{0.375}$
Right ($X \geq 2.4$)	2.5, 3.0	2 N (pure)	$1 - 1^2 = \mathbf{0}$

$$J(X, 2.4) = 46(0.375) + 26(0) = 0.25$$
$$J(X, 2.4) = \frac{4}{6}(0.375) + \frac{2}{6}(0) = \mathbf{0.25}$$
$$\mathbf{0.25} J(X, 2.4) = 64(0.375) + 62(0) = 0.25$$

Result

Split	J
X = 1.35	0.40
X = 2.4	0.25 ✓